Machine Learning for Cybersecurity (5883V/Ü)

1. Organisation, Introduction to Security, Introduction to Machine Learning

Hans P. Reiser | Winter term 2026/27

University of Passau



AI GENERATED IMAGE REPRESENTING MACHINE LEARNING AND CYBER SECURITY

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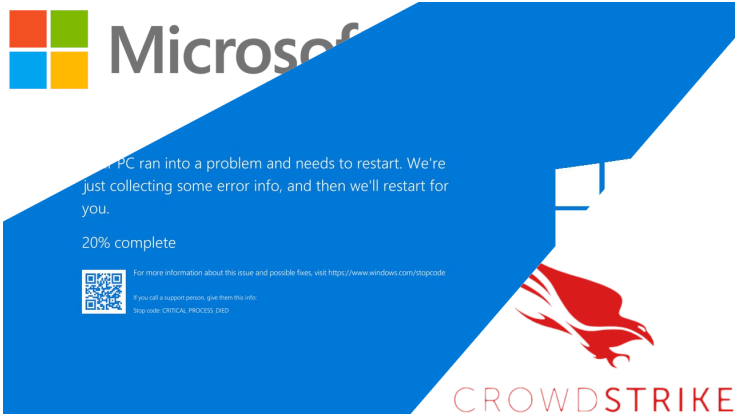
AI GENERATED IMAGE REPRESENTING MACHINE LEARNING AND

What is ML + CS?

“The CrowdStrike Falcon sensor delivers AI and machine learning to protect customer systems”

What is ML + CS?

“The CrowdStrike Falcon sensor delivers AI and machine learning to protect customer systems”



Millions of computers crash, US $10.000.000.000 damage. . .

. . . even without an attack

Frontier models create new attacks

- LLM-written spearphishing: fluent, personalised, cheap at scale
- Agent-automated tool chains: faster attacks with little human in the loop
  - Reconnaissance, exploit attempts, data theft, ...

[Anthropic, 2025; The Register, 2024]

Overview

1. Class organisation

2. Security basics
   - Risk management
   - Terminology
   - Cybersecurity

3. Machine learning basics
   - What machine learning can bring to cybersecurity
   - ML terminology and methodologies
   - ML tasks and cyber security
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Lecture MLCS

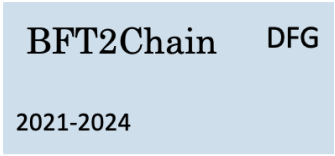
- Instructor: Hans P. Reiser <hr@sec.uni-passau.de>
  - Studied computer science at FAU Erlangen, Germany
  - PhD in computer science from Ulm University, Germany
  - Postdoc and assistant professor at University of Lisbon, Portugal
  - Assistant professor at University of Passau, Germany
  - Visiting professor at Carnegie Mellon University, US, and KIT, Germany
  - since 2022 at Reykjavik University, co-founder of Frostbyte research centre
  - since 7/2026 Full Professor at University of Passau, Germany
    (and part-time professor at RU)

- Research focus
  - Secure and reliable distributed systems, in particular:
  - Intrusion detection, malware analysis, digital forensics
  - Intrusion tolerance – Byzantine fault tolerant (BFT) systems

About me: Research projects

- Intrusion-tolerant and secure distributed systems







- Hypervisor-based monitoring, attack detection, and forensic analysis









Lecture

- Lecture material on Stud.IP
  - News and announcements
  - Lecture slides
  - Exercises (practical assignments)
  - Forum for discussions
- Lecture format
  - Tuesday and Thursday time slots flexibly used for lectures, assignment discussions, and presentations

Literature

Main literature: **research papers** (individual announcements for each week)

Additional recommendations of useful books:

- Clarence Chio and David Freeman: “Machine Learning & Security – Protecting Systems with Data and Algorithms” (O’Reilly) [Chio and Freeman, 2018]
- Chiheb Chebbi: “Mastering Machine Learning for Penetration Testing” (Packt) [Chebbi, 2018]

Class structure

First 3–4 weeks:

- Traditional classes with an introductory part
- Simple introductory exercises

Afterwards:

- Introduction lecture for each new topic
- In class: student presentations of research papers + discussions
- In-class help sessions for assignments
- Presentation of programming-assignment results

Course assessment

Portfolio of

- Paper presentation (∼20%)
- Final project (∼40%)
- Presentation and discussion of final project (∼40%)

Portfolio assessment requires doing all parts, one grade at the end!

Lecture topics overview (+ tentative assignments)

- Spam detection (A1)
- Network intrusion detection (A2, presentations)
- Malware analysis and classification (A3, presentations)
- Cyber Threat Intelligence (presentations)
- Adversarial machine learning (A4)
- ML-based attacks and AI agents in cybersecurity (presentations)
- Attacks on classical classifiers: evasion, poisoning (presentations)
- Attacks on LLMs/agents: prompt injection, jailbreaks, tool abuse, RAG/data poisoning (presentations)
- AI use in digital forensics (presentations)
- Privacy and ML

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What is security?

What is security?

“X is secure” as an isolated statement does not make any sense!
(for multiple reasons):

- A system is secure *against something* (some threat)
  - First task: Analysis of threats and protection goals!
- Security is relative
  - How much effort (time, money) does an attacker need in order to be successful
  - In practice: cost/benefit analysis is necessary
- Security measures need to be understood as tools for risk management

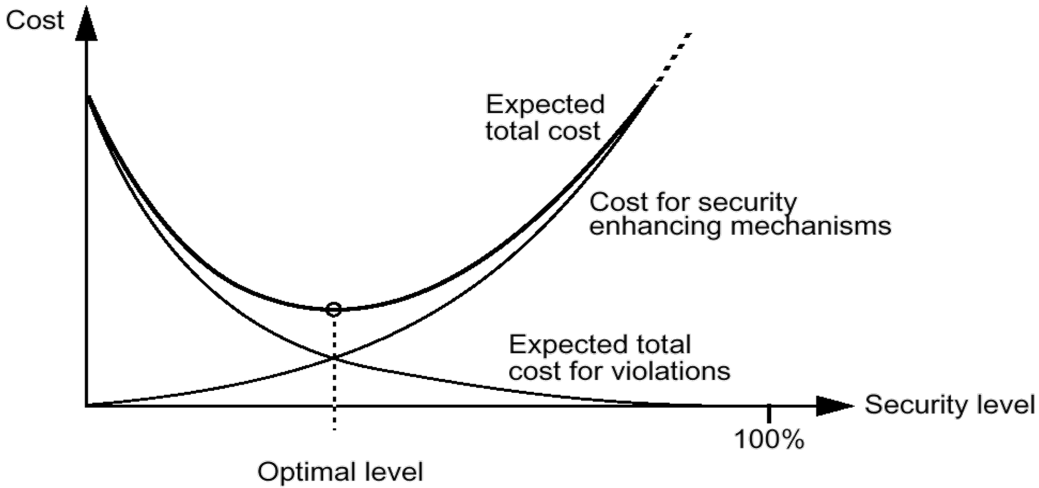
Elements of security

Policy: ***Specifying* security**
What is it supposed to do?

Mechanism: ***Implementing* security**
How does it do it?

Assurance: ***Correctness* of security**
Does it really work?

Cost/benefit analysis (in theory...)



Example: Security goals for mobile phone service

Protection against:

- Eavesdropping of conversation on air
  - Encryption of communication
- Extracting information from signalling channel on air
  - Encryption of communication
- Unauthorised usage
  - Identification and authentication of participant (GSM: IMSI)
- Theft of devices
  - Identification of end devices (IMEI)

Terminology: Elements of security

assets	risks
threats	safeguards
vulnerabilities	residual risk

Terminology: Elements of security

assets

threats

vulnerabilities

risks

safeguards

residual risk

Assets: Everything that has some value, e.g.:

- Information and data
- Hardware and software
- Documents
- Services
- Trust in services
- Reputation of an organisation
- ...

Terminology: Elements of security

assets

threats

vulnerabilities

risks

safeguards

residual risk

Threats: Everything that could harm assets, e.g.:

- Faults
- Misuse and theft
- “Malicious code”
- Hacking
- Sabotage
- Espionage
- . . .

Terminology: Elements of security

assets

threats

vulnerabilities

risks

safeguards

residual risk

Vulnerabilities: Weaknesses that allow a threat to succeed against an asset.

(do not cause damage directly, but may make damage possible):

Examples:

- Insecure communication channels
- Badly trained employees
- Trivial passwords
- Lack of access control
- No backup
- Flaws in software
- ...

Terminology: Elements of security

assets	risks
threats	safeguards
vulnerabilities	residual risk

Safeguards: Measures that reduce threats or vulnerabilities

Examples:

- Access control
- Encryption
- Education of employees
- ...

Terminology: Elements of security

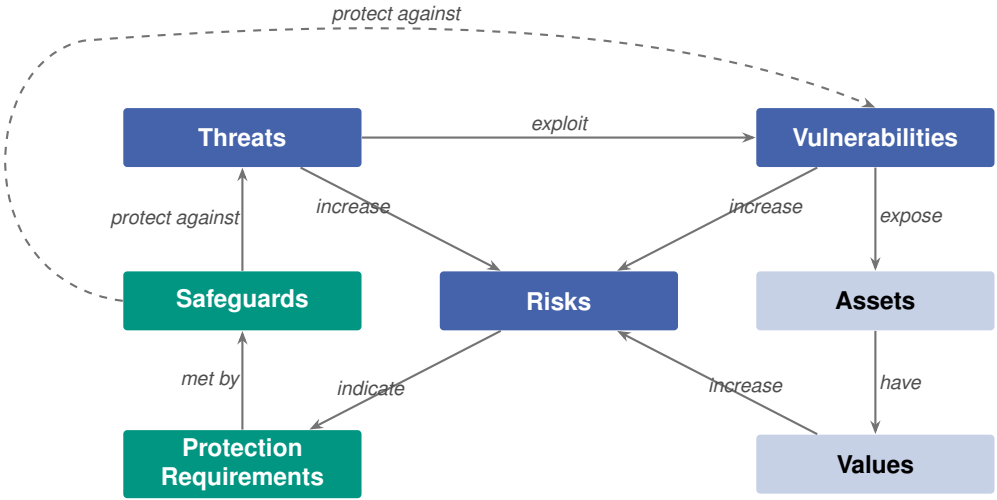
assets	risks
threats	safeguards
vulnerabilities	residual risk

Risk: Function of

- assets
- threats
- vulnerabilities
- safeguards

Note: There is always some residual risk.

Relations between terms



What is IT security/cyber security?

What is IT security/cyber security?

Classical answer

- Cryptology (originally from military applications)

In practice:

- Cryptography (encryption, authentication, watermarking, protocols, PKIs, . . .)
- Software engineering (bug avoidance, verification, evaluation/certification)
- Platform security and security architectures (operating systems, virtual machines, etc.)
- Internet security (protocols, secure web servers, firewalls, online payments, . . .)
- Hardware security (smartcards, secure tokens, etc.)
- . . .

Classification of attackers

(see also: Abraham, Dolan, Double, Stevens: Transaction Security System, IBM Systems Journal, 30(2), 1991)

- **Class I: “Clever Outsiders”.** Intelligent, but lack expert knowledge about the system, only moderately sophisticated equipment, exploit vulnerabilities in existing systems.
- **Class II: “Knowledgeable Insiders”.** Highly educated and have detailed understanding of parts of the system. Potential access to most of the system. Often have access to highly sophisticated equipment.
- **Class III: “Funded Organisations”.** Teams of specialists that may include knowledgeable insiders, backed by great funding resources. They are capable of designing sophisticated attacks and use the most sophisticated equipment.

Basic terminology: C – I – A

- **Confidentiality:** the prevention of the unauthorised disclosure of information;
- **Integrity:** the prevention of the unauthorised modification of information
- **Availability:** the prevention of the unauthorised withholding of information or resources.

(according to ITSEC - the European Information Technology Security Evaluation Criteria)

Confidentiality

Historically, security and secrecy were closely related, and sometimes, security and confidentiality are used as synonyms

- Prevent unauthorised disclosure of information (prevent unauthorised reading/learning)

- Privacy: protection of personal data
- Secrecy: protection of data belonging to an organisation

Integrity

Everything is the way it is supposed to be...

- ITSEC: prevent unauthorised modification of information
- Clark and Wilson: No user of the system, even if authorised, may be permitted to modify data items in such a way that assets or accounting records of the company are lost or corrupted.
- Orange Book (TCSEC): Data Integrity – The state that exists when computerised data is the same as that in the source document and has not been exposed to accidental or malicious alteration or destruction.
- In communications: detection (and correction) of modifications of transmitted data, including both intentional modifications and random transmission errors.

Availability

- ISO 7498-2: the property of being accessible and usable upon demand by an authorised entity
- Denial of Service: The prevention of authorised access of resources or the delaying of time-critical operations

Accountability and non-repudiation

- Accountability: Users should be held responsible for their actions
  - Orange Book: Audit information must be selectively kept and protected so that actions affecting security can be traced to the responsible party
  - System needs to identify and authenticate users, and keep an audit trail of security-relevant events.

- Non-repudiation: related to accountability:
  - Provide unforgeable evidence that a specific action occurred
  - Facets: non-repudiation of origin, non-repudiation of delivery

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What machine learning can bring to cybersecurity

(based on [Forbes Technology Council, 2021])

- AI/ML is being applied in enterprises
  - Decision support, recommendations, process control, etc.
  - Using big-data-driven algorithms, approximating human decision makers
- Systems improved through domain-specific data
  - Iteratively adapting algorithms to model the domain
  - Retail example: Utilising sales data to suggest additional purchases (using correlations of purchases of past customers), potentially increasing sales
- Increasingly, AI/ML integrated creatively into enterprise IT security
  - handling large amounts of data from IT development, operations, network, etc.
- AI/ML cybersecurity applications
  - employ an ML model that learns iteratively
  - mimicking the decision-making processes of humans (trained professionals)

What machine learning can bring to cybersecurity

(based on [Forbes Technology Council, 2021])

- Basic anomaly detection with data science
  - Correlations and standard deviations for real-time identification of out-of-range events
- Fundamental problem: Correlation does not necessarily imply causation
  - Outliers aren't necessarily threats, anomalies can be due to potential unrelated reasons like chance, new applications, or workforce changes
  - Potentially many false positives that need to be investigated manually
  - AI/ML models analyse beyond outliers, evaluating risk levels based on past learning
  - Enables focus on high-risk events, streamlining security efforts by considering history and experience
- Required ingredient: data access
  - AI/ML systems need to assess data (logs, other sources) promptly for threat detection
  - Direct APIs from diverse data sources enhance AI/ML integration for cybersecurity
  - Swift data extraction from logs and sources enables near real-time examination by AI/ML

What machine learning can bring to cybersecurity

Machine learning is not a silver bullet for cybersecurity

Main ML applications in cyber security:

- Network traffic
- Malware
- Application security
- User and process behaviour

Main ML tasks in cyber security:

- Regression / prediction
- Classification
- Clustering



Since Forbes (2021)

- Fast-paced evolution
  - LLMs *on top* of classical detectors: triage, summarisation, querying logs
  - Agents that call tools: SIEM, EDR, tickets, sandboxes, ...
- Old problems remain and new ones appear
  - False positives; correlation $\neq$ causation
  - Hallucination, prompt injection, over-trust in copilots
- A challenge for this class
  - Automating simple tasks with AI does not replace understanding them
  - Without the basics, the complex parts stay out of reach

1 Class organisation

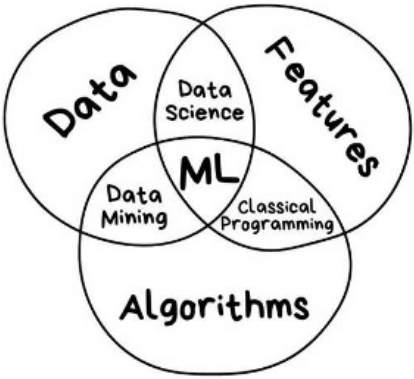
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Distinguishing AI, ML and DL

- Artificial Intelligence (AI): A broad concept, a science of making things “smart”, i.e. making machines perform human tasks
- Machine Learning (ML): An approach to AI that uses a system capable of learning from experience
  - Intended not only for AI, can reduce efforts for simple and difficult tasks
  - ML system recognises patterns by using examples, not by explicitly programming them
  - If a system learns constantly, makes decisions based on data rather than algorithms, can change its behaviour $\Rightarrow$ system is ML
- Deep Learning (DL): A set of techniques for implementing ML
  - Usually based on multi-layered artificial neural networks

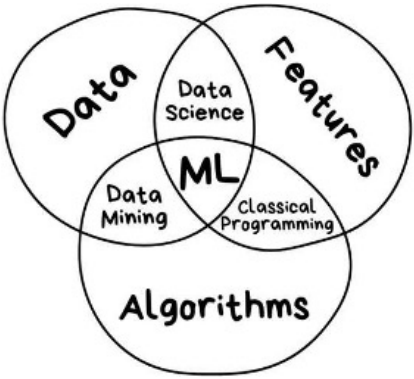
Machine learning tasks



Most common ML tasks:

- Regression (prediction): Predicting a continuous numerical value based on input data.
- Classification: Separating things into different categories/classes
- Clustering: ($\approx$ classification, but unknown classes) Grouping things by their similarity
- Association rule learning: Recommending something based on the previous experience
- Dimensionality reduction (generalisation): Searching common and most important features in multiple examples
- Generative models: Creating something based on the previous knowledge of a distribution

Machine learning tasks



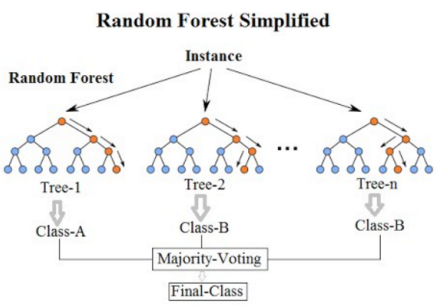
There are different approaches for these tasks . . .

Most common ML tasks:

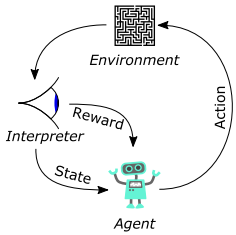
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Approaches to solving ML tasks

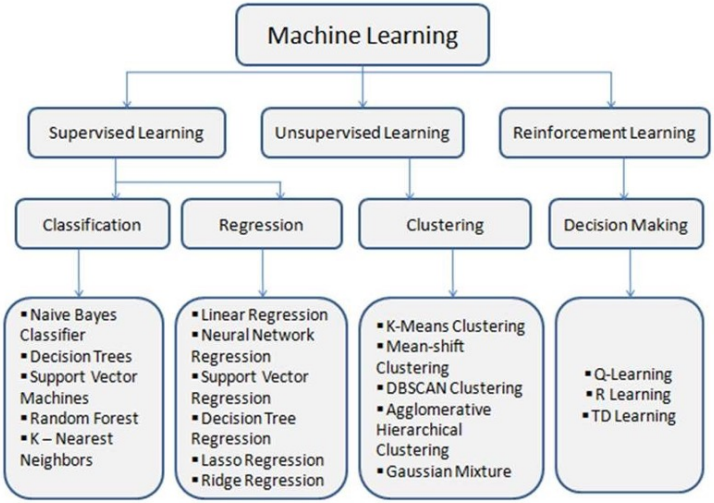
- Supervised learning: “task-driven approach”:
  - Learning with labelled data set (e.g., files + is it malware?)
  - Model makes decision about new data
- Ensemble learning:
  - Extension of supervised learning (combining different simple models to solve task)
- Unsupervised learning: “data-driven approach”:
  - No labelled data; model shall somehow mark data
  - Finding anomalies (generally more powerful, less precise)
- Semi-supervised learning:
  - Combining supervised and unsupervised approaches
- Reinforcement learning: “environment-driven approach”:
  - Model must learn environment properties, behaviour should react to changing environment
- Active learning: Actively selecting which data items to label



Random Forest licensed from Venkata Jagannath under CC BY-SA 4.0



Classifying ML approaches



source: https://medium.com/machine-learning-community/everything-you-need-to-know-about-machine-learning-technology-f7e9b6ce07ce

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Regression (prediction)

- Knowledge from existing data is used to have an idea of new data
- Cyber security example: regression to predict the next user action and detect outliers such as credit card fraud
- Some algorithms for regression:

Machine learning algorithms	Deep learning algorithms
Linear regression	Artificial neural network (ANN)
Polynomial regression	Recurrent neural network (RNN)
Ridge regression	Neural Turing machines (NTM)
Decision trees	Differentiable neural computer (DNC)
Random forest	
Support vector regression (SVR)	

Classification

- Knowledge of existing, labelled data is used to classify new data into known classes
- Cybersecurity example: a spam filter separating spam from other messages
- Some algorithms for classification:

Machine learning algorithms	Deep learning algorithms
Logistic regression (LR)	Artificial neural network (ANN)
K nearest neighbours (KNN)	Convolutional neural network (CNN)
Support vector machine (SVM)	
Kernel SVM	
Decision tree classification	
Random forest classification	

Clustering

- Main difference to classification: classes of data unknown
- Cybersecurity example: Anomaly detection in an intrusion detection system; or grouping malware samples into classes of similar characteristics
- Often used as one step (e.g. to cope with heterogeneous devices in a network, first clustering splits devices into classes, and then within a class, a classifier decides if network traffic is of malicious origin)

Machine learning algorithms	Deep learning algorithms
Bayesian	Self-organised maps (SOM) /
K-means	Kohonen networks
Mixture model (LDA)	
Gaussian mixture model	
Mean-shift	

Association rule learning

- Provide association rules between items based on knowledge about existing data
- Example: Netflix and SoundCloud recommend films or songs
- Cybersecurity example: Incident response (system learns a type of response for a particular incident)
- Cybersecurity example: Risk management (automatically assign risk scores for vulnerabilities based on their description)

Machine learning algorithms	Deep learning algorithms
Apriori	Deep Restricted Boltzmann Machine (RBM)
Eclat	Deep Belief Network (DBN)
FP-Growth	Stacked Autoencoder

Dimensionality Reduction

- Used in complex systems with unlabelled data and many potential features (too many for, e.g., clustering approaches)
- Usually one task in a more complex pipeline
- Cybersecurity example: face recognition

Machine learning algorithms
Principal Component Analysis (PCA)
Singular-value decomposition (SVD)
T-distributed Stochastic Neighbour Embedding (T-SNE)
Linear Discriminant Analysis (LDA)
Latent Semantic Analysis (LSA)
Factor Analysis (FA)
Independent Component Analysis (ICA)
Non-negative Matrix Factorisation (NMF)

Generative Models

- Differ from previous tasks: Simulate actual data based on previous decisions
- Cybersecurity example: offensive tasks, e.g. generate input parameters to test an application for injection vulnerabilities

Machine learning algorithms	Deep learning algorithms
Markov chains	Variational autoencoders
Genetic algorithms	Generative adversarial networks (GAN)
	Boltzmann machines

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Machine learning for network protection

- Different areas and protocols (Ethernet, wireless, SCADA, SDNs, . . .)
- Network protection: Intrusion Detection Systems (IDS) traditional signature-based approaches, some forms of ML used already years ago
- ML in network security: Network Traffic Analytics (NTA) aimed at in-depth analysis of all traffic and each layer, detect attacks and anomalies
- ML examples in this area:
  - Regression to predict network packet parameters, compare observations with learned “normal” behaviour
  - Classification to identify different classes of network attacks (scanning, spoofing, DoS, etc.)
  - Clustering for forensic analysis

Machine learning for endpoint protection

- Every endpoint has its own specifics – but common ML tasks
- Solutions may differ depending on endpoint type (workstation, server, mobile, PLC, IoT device, . . .)
  - e.g. Anti-virus software learning/observing features of executable files or behaviour of running processes

- Some ML examples applied in this area
  - Regression to predict the next system call for executable processes
  - Classification to divide programs into categories such as malware, spyware, ransomware
  - Clustering for malware protection on secure email gateways (e.g., separate legitimate file attachments from outliers)

Machine learning for application security

- Application security can differ: there are web applications, databases, ERP systems, SaaS applications, micro services, etc.
- It is almost impossible to build a universal ML model to deal with all threats effectively. However, we can try to solve some tasks.
- Where to use ML in app security?
  - Web Application Firewalls (WAFs), and Code analysis, both static and dynamic.
- Some ML examples applied to this area:
  - Regression to detect anomalies in HTTP requests (for example, XXE and SSRF attacks and auth bypass);
  - Classification to detect known types of attacks like injections (SQLi, XSS, RCE, etc.);
  - Clustering user activity to detect DDoS attacks and mass exploitation

Machine learning for user behaviour

- Security Information and Event Management (SIEM) systems are able to solve numerous tasks if configured properly (including user behaviour search)
- But they cannot handle all issues that arise from user behaviour. There are domain users, application users, SaaS users, social networks, messengers, and other accounts that should be monitored

- Unlike malware detection focusing on common attacks and the possibility to train a classifier, user behaviour is a complex layer and an unsupervised learning problem
- As a rule, there is no labelled dataset as well as any idea of what to look for – the task of creating a universal algorithm for all types of users is tricky
- Some ML examples applied to this area:
  - Regression to detect anomalies in user actions (e.g., login at an unusual time);
  - Classification to group different users for peer-group analysis;
  - Clustering to separate groups of users and detect outliers.

Machine learning for process behaviour

- In process behaviour it is necessary to know a business process in order to find something anomalous.
- Business processes can differ significantly: we can look for fraud in banking and retail system, or a plant floor in manufacturing. The two are totally different, and they demand a lot of domain knowledge.
- Features are different in all processes; so, the way you represent data to your algorithm is essential to achieve precise and accurate results.
- Some ML examples applied to this area:
  - Regression to predict the next user action and detect outliers such as credit card fraud;
  - Classification to detect known types of fraud;
  - Clustering to compare business processes and detect outliers.

Questions?

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